

# Component Showcase

Living documentation for homework-template

Student Name · Student ID

COURSE 101 · Course Name

Not applicable

Instructor Name · Collaborators: None

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## 1 Cheat sheet

Everything this template offers, on one page. Each entry names the environment or command that produces it; the sections that follow show them in use.

### Structure

`problem` ..... a numbered problem  
`subproblems` ..... (a), (b), (c)  
`solution` ..... worked reasoning  
`answer` ..... the final result  
`\DocFolder` ..... include a folder

### Code

`codebox[lang]{file}` ..... titled plate  
`codeplain[lang]` ..... no title bar  
`\inputcode` ..... a real file  
`terminal` ..... shell transcript  
`\code \filepath` ..... inline

### Callouts

`callout` ..... neutral note  
`tip` ..... what to remember  
`warn` ..... a caveat  
`caution` ..... a mistake to avoid

### Figures

`\HomeworkFigure` ..... an image  
`\HomeworkDiagram` ..... a D2 diagram

### Marks

✓ `\rzcheck`    ✗ `\rzcross`  
⚠ `\rzwarnglyph`    ⓘ `\rzinfoglyph`

### Mathematics

`theorem` `lemma` `proposition`  
`corollary` `definition` `remark`  
`proof` ..... ends with a tombstone

ⓘ **Two rules** Write one problem per file in `fragments/problems/` and number with room to grow — 010, 020, 030. Everything in that folder is included in name order, so a new problem is a new

file and `homework.tex` is never edited.

## 2 Problems and solutions

### Problem 1: A numbered problem

*12 points*

Problem statements may contain prose, displayed mathematics, lists, and references. The optional argument records the available points.

- (a) The first subproblem.
- (b) The second subproblem.

 **Solution** Solutions use a distinct breakable panel and may span multiple pages.

 **Final answer** Use an answer box when the final result should be easy to locate.

### Problem 2: A problem without points

The points argument is optional, but the descriptive title is required.

## 3 Callouts

Four boxes for interrupting the argument. Each carries an icon and a word as well as a colour, so a reader in greyscale still gets which kind it is. The optional argument replaces the label.

 **Note** Neutral context. The default label comes from the environment name.

 **Takeaway** The one sentence you want the grader to remember.

 **Watch the bound** The recurrence only closes for  $n \geq 1$ ; the base case has to be checked separately.

 **A common mistake** Do not assume the loop terminates because it does on your inputs.

**i Custom label** All four take an optional argument. Giving one turns the label into the sentence it introduces, rather than a generic banner.

## 4 Mathematics

### Theorem 1 (Geometric series)

For  $r \neq 1$  and every integer  $n \geq 0$ ,

$$\sum_{k=0}^n r^k = \frac{1 - r^{n+1}}{1 - r}.$$

*Proof.* Multiply the sum by  $1 - r$  and cancel adjacent terms. □

The template includes theorem, lemma, proposition, corollary, definition, and remark environments. Equations, theorems, figures, and tables can be referenced with `\cref`.

## 5 Images and diagrams

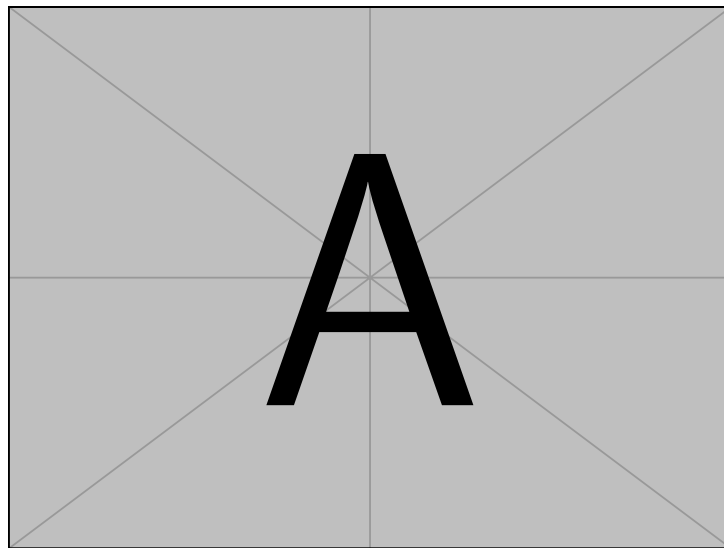


Figure 1: A sample raster-compatible figure included by the TeX distribution.

Every figure takes a visual description as its fourth argument, separate from the caption. The caption is a label for a reader who can see the figure; the description is the figure, in words, for one who cannot. [Figure 1](#) carries both.

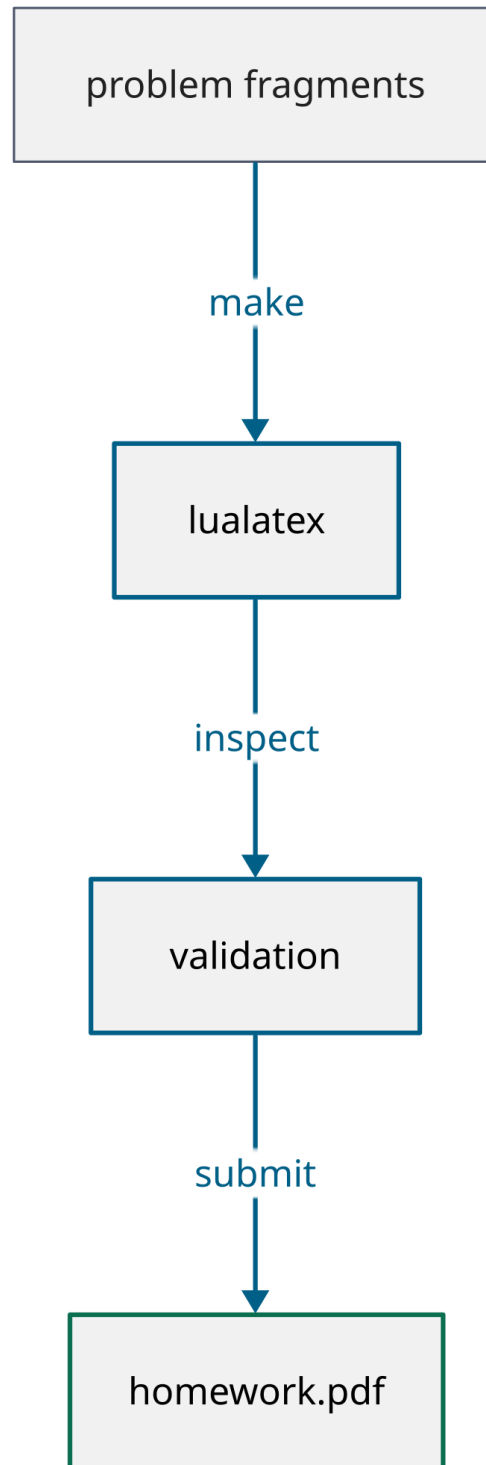


Figure 2: The source-to-submission workflow, drawn in D2.

Diagrams are written in D2 under `assets/diagrams/` and compiled to PDF by `make diagrams`. The generated PDFs stay committed, so editing a diagram needs D2 installed but compiling the document does not.

A diagram needs its description more than a photograph does, and here for two reasons rather than one.

The first is that the labels are not in the text layer at all. The pipeline goes through PostScript on its way to PDF, and that hop outlines the glyphs, so `pdftotext` returns nothing for a diagram. That is a deliberate trade: the PostScript route produces a byte-identical PDF on every run, which is what lets the build assert that a committed diagram still matches its source, and the direct route keeps the text but changes bytes between runs.

The second reason would hold even if the words were there. They would arrive as a loose sequence in drawing order, with no edges, no direction and no shape. Extractable text is not a description.

## 6 Code, tables, and citations

`binary_search.py`

```

1  def binary_search(values, target):
2      low, high = 0, len(values) - 1
3      while low <= high:
4          middle = (low + high) // 2
5          if values[middle] == target:
6              return middle
7          if values[middle] < target:
8              low = middle + 1
9          else:
10             high = middle - 1
11     return None

```

Table 1: Asymptotic costs of common search operations.

| Method        | Prerequisite        | Worst case       |
|---------------|---------------------|------------------|
| Linear search | None                | $\Theta(n)$      |
| Binary search | Sorted input        | $\Theta(\log n)$ |
| Hash lookup   | Suitable hash table | $\Theta(n)$      |

Table 1 uses spacing rather than decorative rules or vertical separators. Bibliographic citations use BibTeX and Natbib; for example, see [Knuth \(1984\)](#).

### 6.1 Other languages

The optional argument is the Pygments lexer, so any language Pygments knows is available. The signature is the slides project's on purpose: a fragment written for one template reads the same in the other.

 `sum_array.c`

```

1  size_t sum_array(const int *values, size_t n) {
2      size_t total = 0;
3      for (size_t i = 0; i < n; i++) {
4          total += values[i]; /* no overflow check, on purpose */
5      }
6      return total;
7  }

```

 `Fib.hs`

```

1  fib :: Int -> Integer
2  fib n = fibs !! n
3      where fibs = 0 : 1 : zipWith (+) fibs (tail fibs)

```

## 6.2 Without a title bar

`codeplain` drops the tab and the line numbers, for an excerpt short enough that neither earns its space.

```

SELECT course, AVG(grade) AS mean
FROM submissions
GROUP BY course
HAVING COUNT(*) > 5;

```

## 6.3 Inline

Set an identifier with `\code`, as in `binary_search(values, target)`, and a path with `\filepath`, as in `fragments/problems/010-example.tex`. Both stay in Ubuntu Mono, so an identifier reads the same inline as it does inside a plate.

## 7 Terminal transcripts

When the answer is what a command printed, quote the session rather than paraphrasing it. The body is verbatim, so nothing inside needs escaping except the arguments of the macros themselves.

 `zsh`

```

➔ arch ~/homework make check
>> homework: LaTeX pass 1
homework.pdf: US Letter, tagged
✓ all 42 pairs meet WCAG 2.1 AA
✗ arch ~/homework make submit

```

✖ permission denied

⚠ 2 problems still marked TODO

The prompt follows the `ginger` zsh theme: a green arrow, then the host, then the directory. `\cmdfail` turns the arrow red *and* changes it to a cross, so a failed command is not distinguished by colour alone. `\TermHost` and `\TermPath` set the two labels; leave them empty and the prompt is just the arrow.

## References

Donald E. Knuth. Literate programming. *The Computer Journal*, 27(2):97–111, 1984. doi: 10.1093/comjnl/27.2.97.